

Artificial Intelligence

Course Information

- **Course Code:** CSC462 Course
- **Title:** Artificial Intelligence
 - **Credit Hours:** 4 (3,1)

Text and Reference Books

- **Textbook:** Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.
- **Reference Book:** Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.



Artificial Intelligence Search Agents Informed search

Topics Covered: LOCAL SEARCH AND
OPTIMIZATION PROBLEMS

Reading Material: AIMA: Ch3

Hill Climbing Algorithm in Artificial Intelligence

- Hill Climbing is a local search algorithm used for optimization problems. It starts with an initial solution and iteratively improves it by making small changes. The goal is to reach the highest peak (for maximization) or lowest valley (for minimization) in the state space landscape.

Key Features

- Greedy Approach: Always moves toward an immediate better solution.
- No Backtracking: Does not revisit previous states.
- Memory Efficient: Only keeps track of the current state.

Hill Climbing Algorithm in Artificial Intelligence

- Always moves in a single direction.
- Terminates when it reaches a peak.
- If new state is better than current state then new state=current state.

Working of Hill Climbing

- **Start** with an initial random solution.
- **Evaluate** the heuristic value of the current state.
- **Generate neighbors** of the current state.
- **Move to the neighbor** with the **highest heuristic value**.
- **Repeat** steps 2–4 until:
 - No better neighbor exists (**local maximum** reached).
 - A goal state is found.
 - A maximum number of steps is reached (to prevent infinite looping).

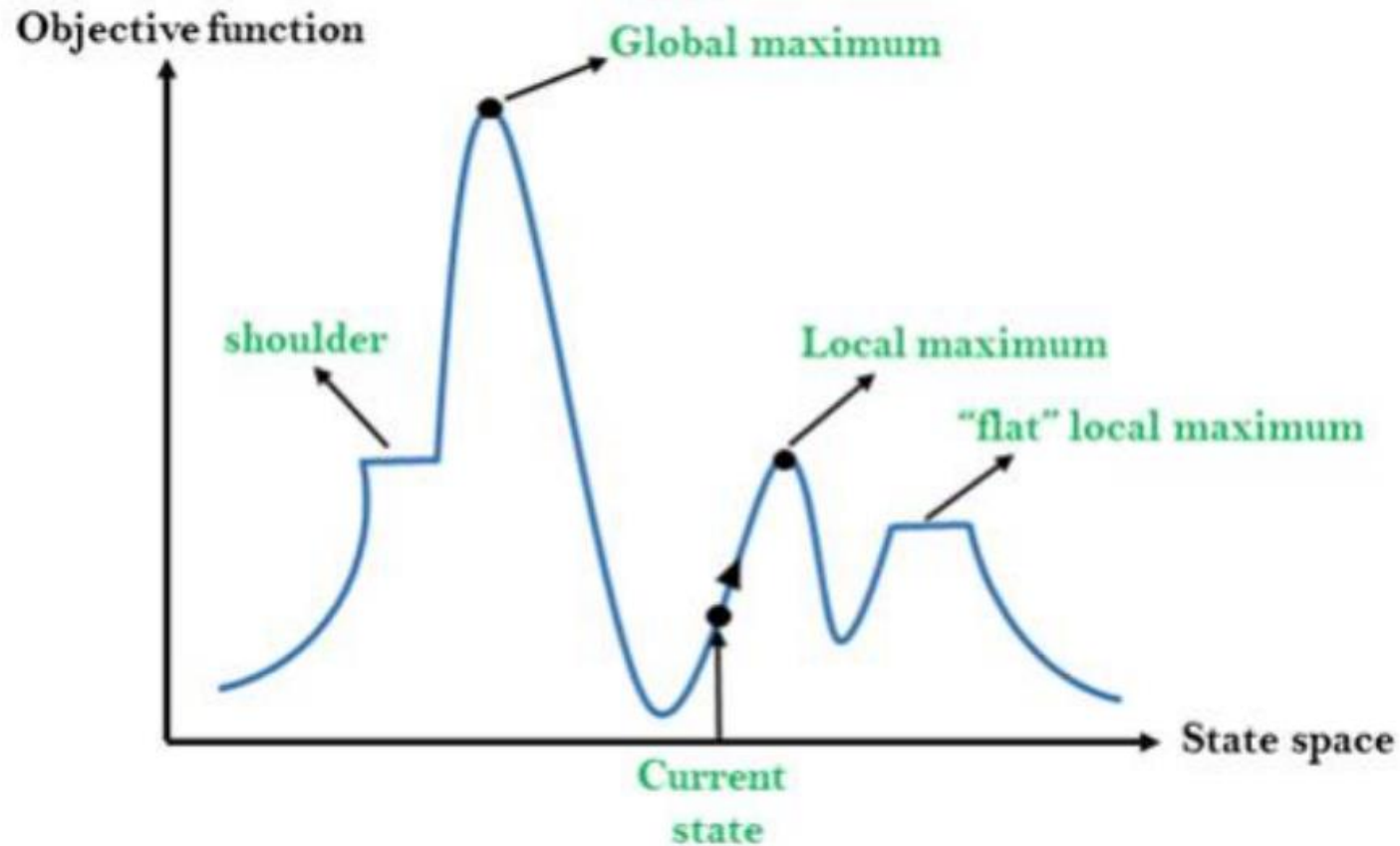
Real-world applications

Application Area	Example	Goal
Robotics	Mars Rover Path Planning	Find safest, shortest path
Job Scheduling	Factory Production Scheduling	Reduce idle time and maximize efficiency
Feature Selection (ML)	Medical Diagnosis Models	Improve model accuracy
Network Routing	Internet Traffic Optimization	Reduce delay and congestion
Game Playing	Chess AI	Make optimal moves
Resource Allocation	Budget Planning	Best use of limited resources
TSP	Delivery Routes (Amazon)	Minimize total distance
Text Summarization (NLP)	News Article Summaries	Extract most important information
Network Security	Intrusion Detection Systems	Detect cyber attacks
Cloud Computing	AWS Resource Allocation	Lower cost, higher performance

State-space Diagram for Hill Climbing:

- The state-space landscape is a graphical representation of the hill-climbing algorithm which is showing a graph between various states of algorithm and Objective function/Cost. On Y-axis we have taken the function which can be an objective function or cost function, and state-space on the x-axis. If the function on Y-axis is cost then, the goal of search is to find the global minimum and local minimum. If the function of Y-axis is Objective function, then the goal of the search is to find the global maximum and local to find the global maximum and local maximum.

State-space Diagram for Hill Climbing:



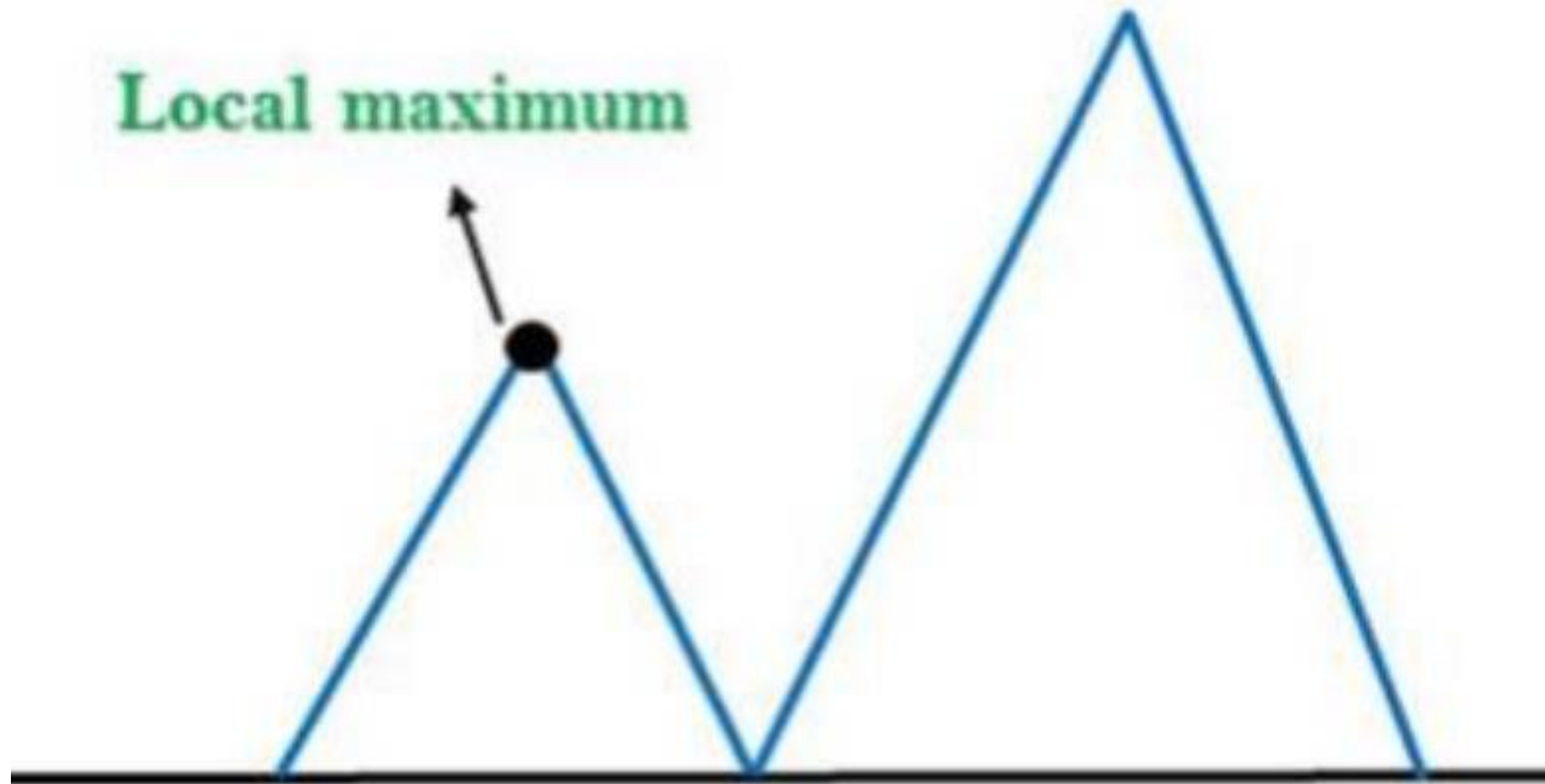
Different regions in the state space landscape:

- **Local Maximum:** Local maximum is a state which is better than its neighbor states, but there is also another state which is higher than it.
- **Global Maximum:** Global maximum is the best possible state of state space landscape. It has the highest value of objective function.
Current state: It is a state in a landscape diagram where an agent is currently present.
- **Flat local maximum:** It is a flat space in the landscape where all the neighbor states of current states have the same value.
- **Shoulder:** It is a plateau region which has an uphill edge.

Problems in Hill Climbing Algorithm

- **Local Maximum:** A local maximum is a peak state in the landscape which is better than each of its neighboring states, but there is another state also present which is higher than the local maximum.
- **Solution:** Backtracking technique can be a solution of the local maximum in state space landscape. Create a list of the promising path so that the algorithm can backtrack the search space and explore other paths as well.

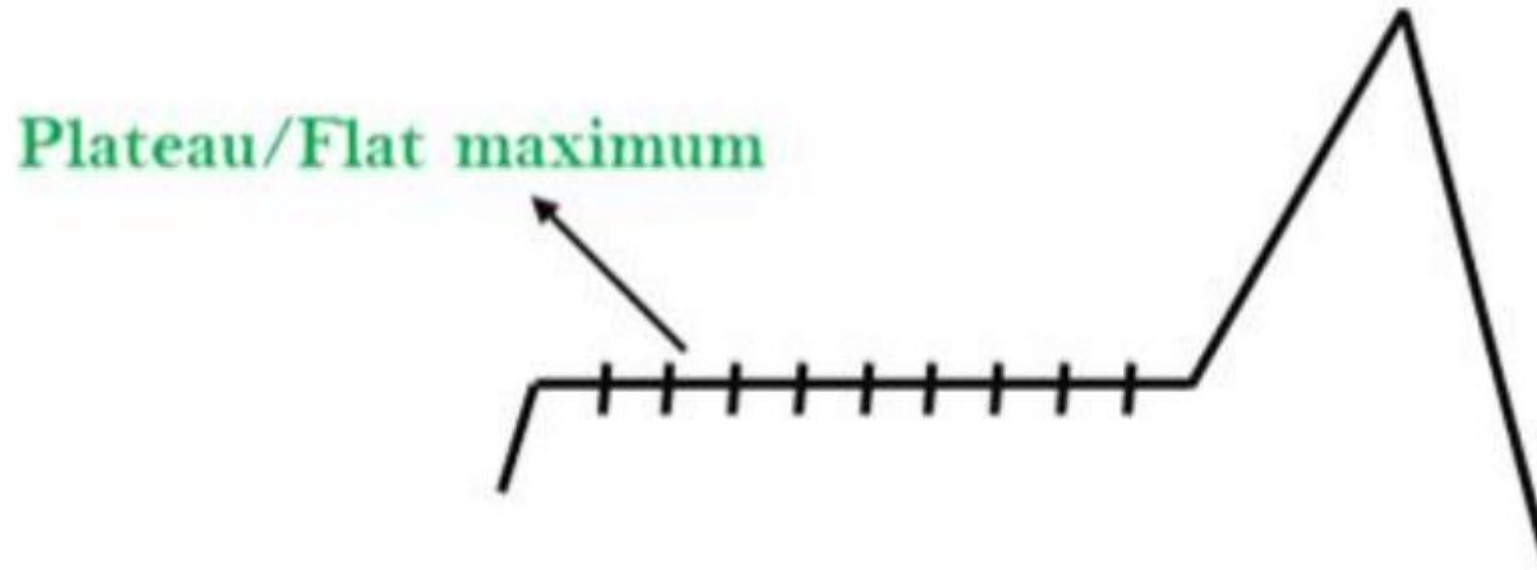
Problems in Hill Climbing Algorithm



Problems in Hill Climbing Algorithm

- Plateau: A plateau is the flat area of the search space in which all the neighbor states of the current state contains the same value, because of this algorithm does not find any best direction to move. A hill-climbing search might be lost in the plateau area.
- Solution: The solution for the plateau is to take big steps or very little steps while searching, to solve the problem. Randomly select a state which is far away from the current state so it is possible that the algorithm could find non-plateau region.

Problems in Hill Climbing Algorithm

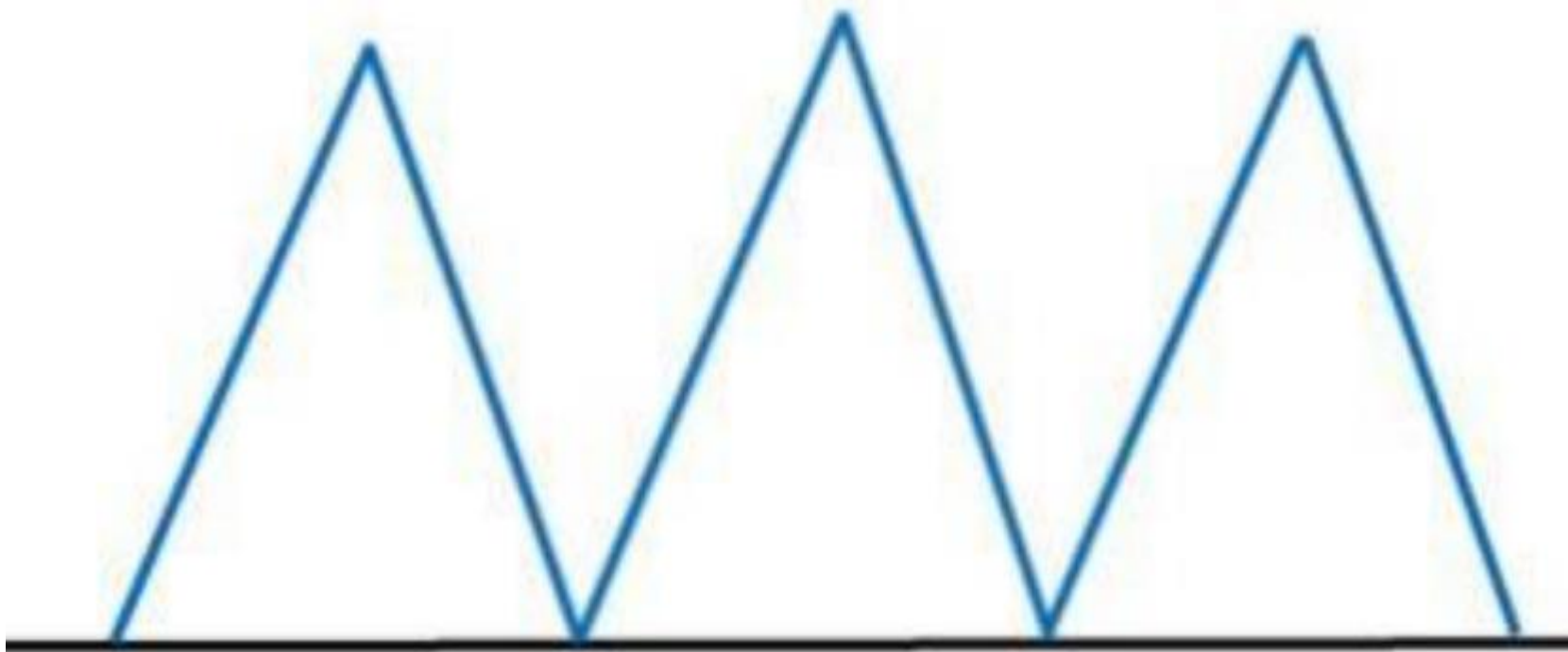


Problems in Hill Climbing Algorithm

- **RIDGES:** A RIDGE IS A SPECIAL FORM OF THE LOCAL MAXIMUM. IT HAS AN AREA WHICH IS HIGHER THAN ITS SURROUNDING AREAS, BUT ITSELF HAS A SLOPE, AND CANNOT BE REACHED IN A SINGLE MOVE.
- **SOLUTION:** WITH THE USE OF BIDIRECTIONAL SEARCH, OR BY MOVING IN DIFFERENT DIRECTIONS, WE CAN IMPROVE THIS PROBLEM.

Problems in Hill Climbing Algorithm

Ridge



Optimization Algorithms

- Optimization Algorithms Objective: Explore optimization algorithms and their applications.
- Implement Hill-Climbing algorithm to solve the 8-queen problem.
- Apply Simulated Annealing algorithm to optimize flight routes for an air travel problem.
- Implement Genetic Algorithms for solving the 8-queen problem.